22/01/2021

Code-A

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***Regd. Office*:** Aakash Tower, 8,Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** **Time : 3.00 Hrs.**

Test - 6

Answer Key

1.	(2)	37.	(4)	73.	(3)	109.	(3)	145.	(3)
2.	(3)	38.	(1)	74.	(1)	110.	(2)	146	(4)
3.	(2)	39.	(3)	75.	(1)	111.	(1)	147.	(3)
4.	(4)	40.	(2)	76.	(1)	112.	(2)	148.	(4)
5.	(4)	41.	(1)	77.	(4)	113.	(4)	149.	(1)
6.	(2)	42.	(3)	78.	(2)	114.	(1)	150.	(1)
7.	(1)	43.	(1)	79.	(3)	115.	(3)	151.	(1)
8.	(4)	44.	(1)	80.	(2)	116.	(4)	152.	(3)
9.	(1)	45.	(1)	81.	(4)	117.	(1)	153.	(2)
10.	(4)	46.	(3)	82.	(4)	118.	(2)	154.	(3)
11.	(3)	47.	(2)	83.	(2)	119.	(2)	155.	(4)
12.	(3)	48.	(1)	84.	(4)	120.	(1)	156.	(4)
13.	(2)	49.	(4)	85.	(4)	121.	(1)	157.	(2)
14.	(1)	50.	(4)	86.	(3)	122.	(4)	158.	(1)
15.	(3)	51.	(4)	87.	(1)	123.	(2)	159.	(4)
16.	(2)	52.	(4)	88.	(4)	124.	(1)	160.	(4)
17.	(2)	53.	(1)	89.	(3)	125.	(3)	161.	(2)
18.	(4)	54.	(2)	90.	(3)	126.	(1)	162.	(3)
19.	(3)	55.	(2)	91.	(1)	127.	(1)	163.	(4)
20.	(1)	56.	(3)	92.	(4)	128.	(2)	164.	(1)
21.	(4)	57.	(4)	93.	(2)	129.	(4)	165.	(1)
22.	(2)	58.	(2)	94.	(3)	130.	(1)	166.	(2)
23.	(4)	59.	(3)	95.	(3)	131.	(2)	167.	(4)
24.	(4)	60.	(2)	96.	(1)	132.	(1)	168.	(1)
25.	(2)	61.	(2)	97.	(4)	133.	(1)	169.	(3)
26.	(1)	62.	(3)	98.	(3)	134.	(2)	170.	(2)
27.	(3)	63.	(2)	99.	(4)	135.	(3)	171.	(3)
28.	(2)	64.	(2)	100.	(2)	136.	(1)	172.	(4)
29.	(2)	65.	(1)	101.	(3)	137.	(4)	173.	(1)
30.	(2)	66.	(4)	102.	(1)	138.	(4)	174.	(4)
31.	(3)	67.	(2)	103.	(1)	139.	(3)	175.	(2)
32.	(2)	68.	(2)	104.	(2)	140.	(1)	176.	(1)
33.	(2)	69.	(1)	105.	(1)	141.	(3)	177.	(4)
34.	(1)	70.	(4)	106.	(3)	142.	(3)	178.	(4)
35.	(3)	71.	(3)	107.	(4)	143.	(2)	179.	(4)
36.	(1)	72.	(3)	108.	(2)	144.	(4)	180.	(2)

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Test - 6

Hints and Solutions

PHYSICS

1. Answer (2)

If charge q is displaced from null point along z-axis, there is restoring force. But displacing it normal to the z-axis (along x and y-axis), the net force takes it away from null point.

2. Answer (3)

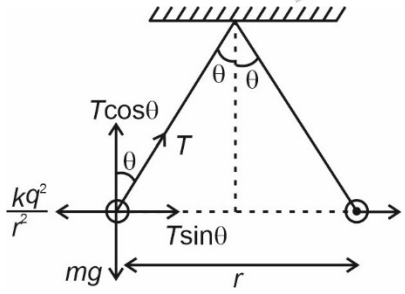
$\because \quad E = \frac{\lambda}{2\pi\varepsilon_0 r}\sin\left(\frac{\theta}{2}\right)$

$E = \frac{\lambda}{2\pi\varepsilon_0 r}\sin\left(\frac{\pi}{4}\right)$

$E = \frac{\lambda}{2\sqrt{2}\,\pi\varepsilon_0 r} = \frac{2\times10^2}{2\sqrt{2}\pi\varepsilon_0\times10^{-2}} = 2.5\times10^{14}$ V/m

3. Answer (2)

At equilibrium the horizontal and vertical component of forces are balanced.



$T\cos\theta = mg$ … (i)

$T\sin\theta = \frac{kq^2}{r^2}$ … (ii)

(ii) / (i)

$\tan\theta = \frac{kq^2}{mgr^2}$

$q = \sqrt{\frac{\tan\theta mgr^2}{k}}$

$q = \sqrt{\tan 30° mgr^2(4\pi\varepsilon_0)}$

$q = \frac{\left[4\pi\varepsilon_0 mgr^2\right]^{1/2}}{\sqrt{3}}$

4. Answer (4)

Initially

$F = \frac{K|q_1||q_2|}{r^2} = \frac{K\times6\times2}{r^2} = \frac{12K}{r^2}$

Finally

$F' = \frac{K\frac{|q_1|}{2}\left|\frac{\frac{q_1}{2}+q_2}{2}\right|}{r^2}$

$F' = \frac{K\times\frac{6}{2}\times\left|\frac{\frac{6}{2}-2}{2}\right|}{r^2}$

$F' = \frac{K\times3\times\frac{1}{2}}{r^2} = \frac{3K}{2r^2}$

$F' = \frac{F}{8}$

5. Answer (4)

Electric flux $\phi = \vec{E}\cdot\vec{A}$

$\phi = \left(3\hat{i} + 4\hat{j} + \hat{k}\right) \cdot (10^{-3}\,\hat{j})$

$\phi = 4 \times 10^{-3}$ Nm^2/C

6. Answer (2)

Flux from a cube $\phi = \dfrac{q}{\varepsilon_0}$

$\therefore \quad [\phi]_{\text{square}} = \dfrac{q}{6\varepsilon_0} = \dfrac{2}{6\varepsilon_0} = \dfrac{1}{3\varepsilon_0}$

7. Answer (1)

The net electric field between the plates

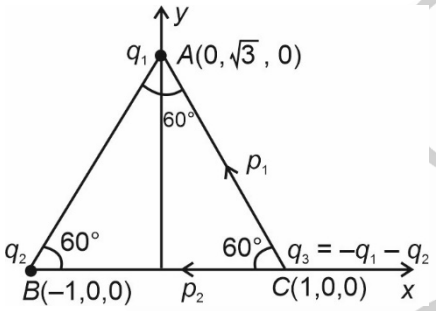
$E = \dfrac{\sigma}{2\varepsilon_0} + \dfrac{\sigma}{2\varepsilon_0} = \dfrac{\sigma}{\varepsilon_0}$

$\sigma = E\varepsilon_0$

$\sigma = 1.2 \times 10^{-10} \times 8.85 \times 10^{-12} = 10.6 \times 10^{-22}$

$\sigma \approx 1.1 \times 10^{-21}$ C/m^2

8. Answer (4)



$\because \quad AB = BC = CA = 2 \times 10^{-3}$ m

$\angle A = \angle B = \angle C = 60°$

$CA = CB = CC = 60°$

$\overrightarrow{p_1} = q_1\,\overrightarrow{CA}$

$\overrightarrow{p_2} = q_2\,\overrightarrow{CB}$

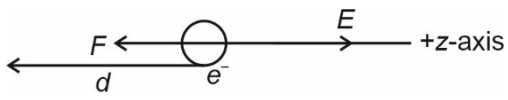
$p_{\text{net}} = \left[p_1^2 + p_2^2 + 2p_1p_2\cos 60°\right]^{1/2}$

$p_{\text{net}} = \left[\begin{array}{c}(q_1 \times 2 \times 10^{-3})^2 + (q_2 \times 2 \times 10^{-3})^2 \\ 2 \times q_1 \times q_2 \times 4 \times 10^{-6} \times \dfrac{1}{2}\end{array}\right]^{1/2}$

$p_{\text{net}} = 2\sqrt{21} \times 10^{-9}$ Cm

$p_{\text{net}} = 9.2 \times 10^{-9}$ Cm

9. Answer (1)



Work done by electric force

$W_E = Fd = (qE)d \left[\because \ \theta = 0°\right]$

$\because \quad \Delta U = -W_E$

If $W_E > 0$ then $\Delta U < 0$

$\because \quad U_f < U_i$

Hence potential energy decreases.

10. Answer (4)

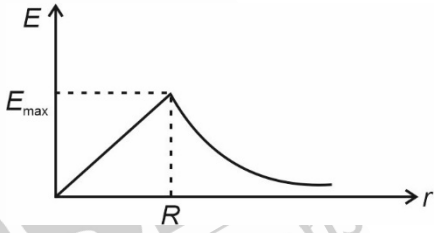
$a = \dfrac{qE}{m}$ and $v^2 = u^2 + 2as$

$v^2 = 2\left(\dfrac{qE}{m}\right)s$

$K = \dfrac{m}{2}v^2 = qEs$

$\therefore \quad \dfrac{K_\alpha}{K_p} = \dfrac{q_\alpha}{q_P} = 2$

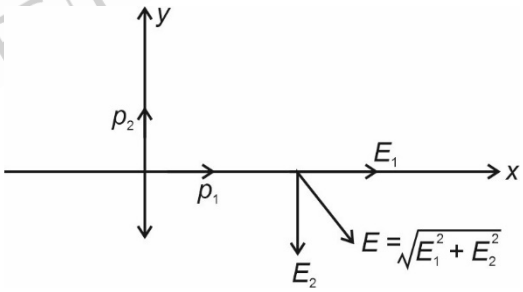
11. Answer (3)



$E \propto r \ [r < R]$

$E = \dfrac{1}{r^2} [r > R]$

12. Answer (3)



$E_1 = E_{\text{axis}} = \dfrac{K\,2p}{r^3}$ and $E_2 = \dfrac{K_p}{r^3}$

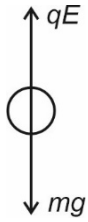
$E = \sqrt{\left(\dfrac{K\,2p}{r^3}\right)^2 + \left(\dfrac{K_p}{r^3}\right)^2} = \dfrac{K\sqrt{5}\,p}{r^3}$

$E = 9 \times 10^9 \sqrt{5} \times \dfrac{\sqrt{5} \times 10^{-8}}{1} = 450$ V/m

13. Answer (2)

$V_{\text{surface}} = \dfrac{kq}{R}$ = constant [at $r < R$]

14. Answer (1)



$mg = qE$

$mg = 16eE$

$\rho\frac{4}{3}\pi r^3 g = 16\ eE$

$r = \left[\frac{48eE}{4\pi\rho g}\right]^{1/3}$

$r = \left[\frac{12eE}{\pi\rho g}\right]^{1/3}$

15. Answer (3)

At unstable equilibrium position $U_{max} = pE$ and $K_{min} = 0$

At stable equilibrium position

$U_{min} = -pE$ and $K_{max} = \frac{I}{2}\omega^2$

By the conservation of energy

$K_{max} + U_{min} = K_{min} + U_{max}$

$\frac{I}{2}\omega^2 - pE = pE$

$\omega = \sqrt{\frac{4pE}{I}}$, $p = qL$ and $I = 2m\left(\frac{L}{2}\right)^2$

$\omega = \sqrt{\frac{4qLE}{mL^2/2}} = \sqrt{\frac{8qE}{mL}}$

16. Answer (2)

Electric field $E = \frac{V}{d} = \frac{10^3}{10^{-2}} = 10^5$ V/m

From equation of trajectory, deflection is

$y = \left[\frac{qE}{2mv^2}\right]L^2$

$y = \frac{qEL^2}{4K}\left[K = \frac{m}{2}v^2\right]$

$y = \frac{e\times 10^5 \times (2\times 10^{-2})^2}{4\times 1500\ \text{eV}}$

y = 6.7 mm

17. Answer (2)

If final charges are q_1 and q_2 and potential V_1 and V_2 respectively.

$V_1 = V_2$

$\frac{Kq_1}{R_1} = \frac{Kq_2}{R_2}$

$\frac{q_1}{q_2} = \frac{R_1}{R_2} = \frac{4}{8} = \frac{1}{2}$

$q_1 = \frac{1}{(2+1)}\times q_{net} = \frac{20}{3}\text{nC}[q_1 + q_2 = 20]$

$q_2 = \frac{2}{(2+1)}\times q_{net} = \frac{40}{3}\text{nC}$

$V_2 = \frac{Kq_2}{R_2} = \frac{9\times 10^4 \times \frac{40}{3}\times 10^{-9}}{8\times 10^{-2}}$

$V_2 = 1.5 \times 10^3$ V

18. Answer (4)

Loss of energy $\Delta U = \frac{C_1C_2}{2(C_1+C_2)}(V_1+V_2)^2$

$U_i = \frac{C_1V_1^2}{2} + \frac{C_2V_2^2}{2}$

$\because\ V_1 = V_2 = V$

$\therefore\ \Delta U = \frac{4C_1C_2V^2}{2(C_1+C_2)}$ and $U_i = \left(\frac{C_1+C_2}{2}\right)V^2$

$\frac{\Delta U}{U_i} = \frac{4C_1C_2}{(C_1+C_2)^2} = \frac{4\times 4\times 8}{(4+8)^2} = \frac{8}{9}$

$\frac{\Delta U}{U_i} = 89\%$

19. Answer (3)

$\because\ E = \frac{\sigma}{\varepsilon_0} = \frac{\sigma}{A\varepsilon_0}$

Electric field is independent of separation between the plates.

20. Answer (1)

$U = \frac{(C_1+C_2)}{2}V^2$, $C_1 = KC_1$ and $C_2 = KC_2$

$U = \frac{K\cdot(C_1+C_2)V^2}{2}$

$U = \frac{4\times(4+4)\times(100)^2}{2}$

U = 0.16 J

21. Answer (4)

$V = \frac{q}{C}$ and $C = \frac{2\pi\varepsilon_0 \ell}{\ell n\left[\frac{r_2}{r_1}\right]}$

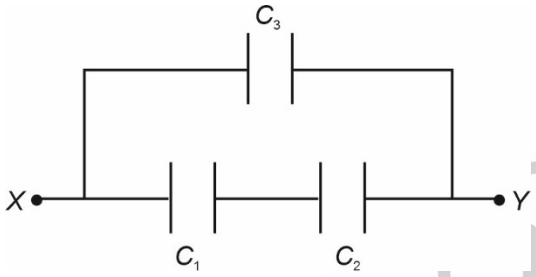
$C = \frac{2\pi\varepsilon_0 \times 0.1}{\ell n\,[2]}$

$V = \frac{q}{C}$

$V = \frac{2\times10^{-6} \times \ell n[2]}{2\pi\varepsilon_0 \times 0.1}$

$V = 2.5 \times 10^5$ V

22. Answer (2)



$C_{XY} = \frac{C_1C_2}{C_1 + C_2} + C_3$

$C_{XY} = \left[\frac{K \times K}{K + K} + 2K\right]C$

$C_{XY} = \frac{5KC}{2}$

$\because \quad C_{XY} = 4C$

$\therefore \quad \frac{5KC}{2} = 4C$

$K = \frac{8}{5}$

23. Answer (4)

Energy stored in capacitor $U = \frac{C_{net} \cdot V^2}{2}$

Work done by battery $W = 2U = C_{net}\, V^2$

$W = (4 + 2) \times (6)^2$

$W = 216\ \mu J$

24. Answer (4)

Potential energy of a system of two charges in an external field.

$U = q_1V_1 + q_2V_2 + \frac{q_1q_2}{4\pi\varepsilon_0 r}$

$U = 2\times10^{-6}V_1 - 2\times10^{-6}V_2 - \frac{2\times2\times10^{-12}}{4\pi\varepsilon_0 \times 0.1}$

$U = 2\times10^{-6}(V_1 - V_2) - \frac{4\times10^{-11}}{4\pi\varepsilon_0}$

$U = 2\times10^{-6}(Ed.) - 9\times10^9 \times 4\times10^{-11}$

$U = 2\times10^{-6} \times 2\times10^4 \times 0.1 - 36\times10^{-2}$

$U = 0.4\times10^{-2} - 36\times10^{-2}$

$U = -356$ mJ

25. Answer (2)

$V_1 = V_2$

$\frac{Kq_1}{R_1} = \frac{Kq_2}{R_2}$

$\frac{q_1}{q_2} = \frac{R_1}{R_2}$ … (i)

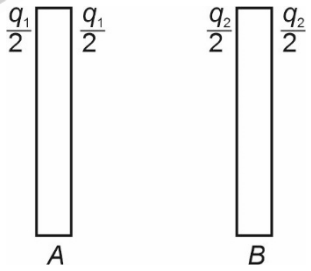
$E = \frac{Kq}{R^2}$

$\frac{E_1}{E_2} = \frac{q_1}{q_2} \times \left(\frac{R_2}{R_1}\right)^2$

From (i)

$\frac{E_1}{E_2} = \frac{R_2}{R_1} = \frac{5}{2}$

26. Answer (1)



Electric field due to plate *A* on plate *B*

$E = \frac{\sigma}{\varepsilon_0}$

$E = \frac{q_1/2}{A\varepsilon_0} = \frac{q_1}{2A\varepsilon_0}$

$\therefore$ Force acting on plate *B*

$F = q_2E$

$F = \frac{q_1q_2}{2A\varepsilon_0}$

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$$F = \frac{q \times \left(\frac{q}{2}\right)}{2A\varepsilon_0}$$

$$F = \frac{q^2}{4A\varepsilon_0}$$

27. Answer (3)

Capacitance of the spherical capacitor

$$C = \frac{4\pi\varepsilon_0 R_1 R_2}{(R_2 - R_1)} \text{ & } V = \frac{q}{C}$$

$$C = \frac{4\pi\varepsilon_0 R \times \frac{3R}{2}}{\left(\frac{3R}{2} - R\right)} = 12\pi\varepsilon_0 R$$

For medium

$C = 12\pi K\varepsilon_0 R$

$C = 12 \times 2\pi\varepsilon_0 R = 24\pi\varepsilon_0 R$

$$V = \frac{q}{C} = \frac{q}{24\pi\varepsilon_0 R}$$

28. Answer (2)

Potential at the centre $V_C = \frac{3}{2}V_s$ = maximum

29. Answer (2)

$$U = \frac{Kq_1q_2}{r}$$

$\because$ $q_1q_2 > 0$, hence if separation is decreased then potential energy increases.

30. Answer (2)

$$\vec{E} = -\frac{dV}{dx}\hat{i} - \frac{dV}{dy}\hat{j} - \frac{dV}{dz}\hat{k}$$

31. Answer (3)

$$w = q[V_B - V_A] = -2\left[\frac{K \times 8}{4} - \frac{K \times 8}{4}\right]$$

$w = 0$

32. Answer (2)

- Force acting on negative charge is directed opposite to the electric field.
- Electric potential decreases in the direction of electric field.

33. Answer (2)

Electric field between the plates

$E_{net} = E + E = 2E$

Energy density

$$u = \frac{\varepsilon_0}{2}E_{net}^2 = \frac{\varepsilon_0}{2}(2E)^2$$

$$u = 2\varepsilon_0 E^2$$

34. Answer (1)

$n \times 200 = 1000$

$n = 5 \Rightarrow$ [5 capacitors should be connected in series in one branch]

$\left(m \times \frac{2}{5}\right) = 4 \Rightarrow m = 10 \Rightarrow$ [10 parallel branches]

Total no. of capacitors $N = mn$

$N = 10 \times 5 = 50$

35. Answer (3)

$K = qV$

$$\frac{m}{2}v^2 = qV$$

$$V = \frac{mv^2}{2q} = \frac{10 \times 10^{-3} \times 25}{2 \times 1 \times 10^{-3}}$$

$V = 125$ V

36. Answer (1)

Potential at the centre of non-conducting sphere

$V_C = \frac{3}{2}V_s$ [V_s is the potential at the surface]

$$V_C = \frac{3}{2} \times 50$$

$V_C = 75$ V

37. Answer (4)

Energy stored in conductor

$$U = \int dw = \int V dq$$

$$U = \int \frac{Kq}{R} dq$$

$$U = \frac{Kq^2}{2R} = \frac{9 \times 10^9 \times (2 \times 10^{-6})^2}{2 \times 0.1}$$

$U = 0.18$ J

38. Answer (1)

$V_{conductor} = V_{centre} = V_{+q} + V_{induced}$

$$V_{conductor} = \frac{Kq}{2R} + 0$$

If q is positive then potential of the conductor will be positive.

39. Answer (3)

$$V_A - V_B = -\int_{B(2,2,0)}^{A(1,1,0)} (\hat{i}+2\hat{j})\cdot(dx\hat{i}+dy\hat{j})$$

$$V_A - V_B = -[x+2y]_{(2,2,0)}^{(1,1,0)}$$

$$V_A - V_B = -[(1-2)+2(1-2)]$$

$$V_A - V_B = -[-1-2] = 3\,V$$

40. Answer (2)

$q = KC_0V = 4 \times 40 \times 10^{-6} \times 5 = 800\ \mu C$

Induced charge $q_{ind.} = q\left[1-\frac{1}{K}\right]$

$$q_{ind.} = 800\left[1-\frac{1}{4}\right]$$

$$q_{ind.} = 600\,\mu C$$

41. Answer (1)

Potential difference is independent of the charge on the outer shell.

42. Answer (3)

Electric potential due to short dipole

$$V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\vec{r}}{r^3}$$

$$V = \frac{1}{4\pi\varepsilon_0}\frac{(p\hat{i})\cdot(-a\hat{i})}{a^3}$$

$$V = \frac{-p}{4\pi\varepsilon_0 a^2}$$

43. Answer (1)

Radial electric field lines are due to the point charge, hence equipotential surface will be spherical.

44. Answer (1)

Electric force acting on the positive charge will be along the electric field hence it will move along the uniform electric field or straight line.

45. Answer (1)

Kinetic energy $K = qV$

$$\frac{mv^2}{2} = qV$$

$$v = \sqrt{\frac{2qV}{m}} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{q_1}{q_2}\times\frac{m_2}{m_1}} = \sqrt{\frac{q}{2q}\frac{4m}{m}}$$

Hence $\frac{v_1}{v_2} = \sqrt{2}$

CHEMISTRY

46. Answer (3)

Concentration term which does not has volume term is independent of temperature.

47. Answer (2)

$$\text{Molality} = \frac{x_{solute}\times 1000}{x_{solvent}\times \text{molar mass of solvent}}$$

$$= \frac{0.2\times 1000}{0.8\times 18} = 13.89\ m$$

48. Answer (1)

As per Henry's law, $p = K_H X_{O_2}$

$$0.5\,atm = 3.5\times10^4\,atm\left(\frac{n_{O_2}}{n_{O_2}+n_{H_2O}}\right)$$

$$0.5 = 3.5\times10^4\times\frac{n_{O_2}}{10}\left[\because n_{O_2}+n_{H_2O}\simeq n_{H_2O}\ \text{in dilute solution}\right]$$

$$n_{O_2} = \frac{5\times10^{-4}}{3.5} = 1.43\times10^{-4} = 0.143\ \text{mmol}$$

49. Answer (4)

Molal elevation constant (K_b) depends on nature of solvent.

50. Answer (4)

Ethyl alcohol and water forms positive deviation solution, for which ΔH_{mix}, ΔS_{mix}, $\Delta V_{mix} > 0$ and $\Delta G_{mix} < 0$.

51. Answer (4)

With increase in number of particles, boiling point of solution increases.

52. Answer (4)

$$P_A = P_A^o\ X_A = 400\times\frac{2}{7} = 114.3\ \text{mmHg}$$

53. Answer (1)

$\Delta T_f = i\,K_f m$ $(i = 1 + 3 \times 0.4 = 1 + 1.2 = 2.2)$

$= 2.2 \times 1.86 \times 0.2 = 0.82$ K

54. Answer (2)

For Isotonic solution

$\pi_1 = \pi_2$

Na_2SO_4 unknown

$(iCRT)_{Na_2SO_4} = (iCRT)$

Unknown

$3 \times 0.25 \times RT = \frac{1 \times 7.5 \times 10}{M} \times RT$

$M = \frac{7.50 \times 10}{0.25 \times 3} = \frac{7.5 \times 1000}{25 \times 3}$

$= \frac{7500}{75} = 100 \text{ g mol}^{-1}$

55. Answer (2)

$Na_2SO_4 \rightarrow 2Na^+ + SO_4^{2-}$

$1 - 0.6 \quad 2 \times 0.6 \quad 0.6$

$= 0.4 \quad = 1.2$

$\therefore i = 0.4 + 1.2 + 0.6 = 2.2$

56. Answer (3)

Solution of negative deviation forms high boiling azeotrope.

57. Answer (4)

Solute which can generate more number of particles on ionization, will have large value of van't Hoff factor.

58. Answer (2)

$\underset{1-\alpha}{2CH_3COOH} \rightleftharpoons \underset{\alpha/2}{(CH_3COOH)_2}$

$i = 1 - \alpha + \frac{\alpha}{2} = 1 - \frac{\alpha}{2} = 1 - \frac{0.75}{2} = 0.625$

$i = \frac{M_{theo}}{M_{obs}} \Rightarrow M_{obs} = \frac{M_{theo}}{i}$

$= \frac{60}{0.625} = 96$

59. Answer (3)

$2 KI + HgI_2 \longrightarrow K_2 [HgI_4]$

So, on addition of HgI_2, no of particles will not increase, so boiling point will not increase.

60. Answer (2)

0.9% $\frac{w}{V}$ NaCl solution (Normal saline solution) has same osmotic pressure as of fluid inside the blood cell.

61. Answer (2)

$2H^+(aq) + 2e^- \longrightarrow H_2(g)$

$E = \frac{0 - 0.0591}{2} \log\left(\frac{pH_2}{[H^+]^2}\right) = \frac{-0.0591}{2} \log\left(\frac{1}{[H^+]^2}\right)$

$= + \frac{0.0591}{2} \log [H^+]^2$

$= + \frac{0.0591}{2} \log (10^{-1})^2 \quad \left[\because pH = 1 \; [H^+] = 10^{-1}\right]$

$= - 0.0591$ V

62. Answer (3)

$E^o_{cell} = \frac{0.0591}{2} \log K_{eq}$

$0.591 = \frac{0.0591}{2} \log K_{eq}$

$0.591 \times 2 = 0.0591 \log K_{eq}$

$20 = \log K_{eq} \Rightarrow K_{eq} = 10^{20}$

63. Answer (2)

According to Faraday's first law, $w = Z I t$

$w = \frac{Eit}{96500} = \frac{108 \times 30 \times 0.965}{96500}$

$= \frac{108 \times 30 \times 965}{96500 \times 1000}$

$= \frac{108 \times 300}{105} = 0.324 \text{ g}$

64. Answer (2)

$CuSO_4(aq) \longrightarrow Cu^{2+}(aq) + SO_4^{2-}(aq)$

Cathode : $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

Anode : $Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$

65. Answer (1)

1000 ml (1 L) of 1 M $CuSO_4$ = 1 mole of Cu^{2+}

According to reaction $Cu^{2+} + 2e^- \rightarrow Cu$

2 F convert 1 mole of Cu^{2+} to Cu.

So $[Cu^{+2}]$ or $[CuSO_4] = 0$ (after electrolysis)

66. Answer (4)

Using Kohlrausch's law

$$\Lambda_m^o(H_3PO_4) = \Lambda_m^o(Na_3PO_4) + 3\times\Lambda_m^o(HCl) - 3\Lambda_m^o(NaCl)$$

$$= x + 3y - 3z$$

67. Answer (2)

More the standard reduction potential, lesser the tendency to oxidise and weaker will be reducing power and vice-versa.

68. Answer (2)

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$$

69. Answer (1)

$$\Lambda_m = \frac{k\times 1000}{C} S\,cm^2mol^{-1}$$

$$= \frac{\left(\frac{\ell}{a}\times\frac{1}{R}\right)\times 1000}{C} \quad \left(\because K = \frac{1}{R}\times\frac{\ell}{a}\right)$$

$$= \frac{1.2\times\frac{1}{250}\times 1000}{0.01}$$

$= 480\ S\ cm^2mol^{-1}$

70. Answer (4)

(i) $A^{2+} + e^- \longrightarrow A^+$ $\quad \Delta G_1^o = -Fx_1$

(ii) $A^{2+} + 2e^- \longrightarrow A$ $\quad \Delta G_2^o = -2Fx_2$

(ii)…(i)

$A^+ + e^- \longrightarrow A$ $\quad \Delta G^\circ = Fx_1 - 2Fx_2$

$= -F(2x_2 - x_1)$

$E^o = 2x_2 - x_1 \left(\because \Delta G^\circ = -nFE^\circ\right)$

71. Answer (3)

For hydrogen concentration cell,

$$E_{cell} = 0.0591\log\frac{[H^+]_{cathode}}{[H^+]_{anode}}$$

If $[H^+]_c > [H^+]_a$ then $E_{cell} > 0$

and cell reaction will be spontaneous.

72. Answer (3)

Smaller the cation, more will be the hydration and so lesser will be the ionic mobility.

73. Answer (3)

During discharging of lead storage battery, Pb and PbO_2 both converted into $PbSO_4$.

74. Answer (1)

With increase in dilution conductivity of strong electryolyte decreases.

75. Answer (1)

Generally, conductors having high value of conductivity than insulators and semiconductors.

76. Answer (1)

SiO_2 (Quartz) is a three dimensional network solid.

77. Answer (4)

Triclinic crystal system has only primitive as possible variation.

78. Answer (2)

No. of atoms $A = 8\times\frac{1}{8} = 1$

No. of atoms $B = 6\times\frac{1}{2} = 3$

No. of atoms $C = 12\times\frac{1}{4} = 3$

Possible formula : AB_3C_3

79. Answer (3)

In a closed packed system of N atoms, no. of T.V and O.V. respectively are 2 N and N.

80. Answer (2)

$$\text{P.F. for bcc} = \frac{\sqrt{3}\pi}{8} = 0.68$$

81. Answer (4)

- When NaCl is dopped with $SrCl_2$, so for every 1 Sr^{2+} ion added 1 cation vacancy developed.
- So, No. of cation vacancies

  $= 10^{-3}\times 10^{-2}\times N_A$

  $= 10^{-5}\times 6\times 10^{23} = 6\times 10^{18}$

82. Answer (4)

MnO is an example of antiferromagnetic substance.

83. Answer (2)

Schottky defect is shown by ionic substances in which the cation and anion are of almost similar size.

84. Answer (4)

For simple cubic unit cell,

$a = 2r$

85. Answer (4)

In AB (NaCl type)

Closest distance between opposite ions = $\frac{a}{2}$

$= \frac{400}{2} = 200$ pm

86. Answer (3)

In fcc (ABCABC…), coordination number of atoms is 12.

87. Answer (1)

$d_{\text{unit cell}} = \frac{Z\,M}{N_A a^3}$

$\frac{d_{sc}}{d_{bcc}} = \frac{\left(\frac{Z}{a^3}\right)_{sc}}{\left(\frac{Z}{a^3}\right)_{bcc}} = \frac{\frac{1}{20^3}}{\frac{2}{40^3}} = 4:1$

88. Answer (4)

CsCl is a bcc unit cell.

89. Answer (3)

$\because$ 4 molecules are in 1 unit cell.

($\because$ It is NaCl type)

$\therefore$ $2N_A$ molecules will be in $\frac{2N_A}{4} = \frac{N_A}{2}$ unit cells

90. Answer (3)

No. of A atoms = 4

No. of B atoms = 4

Formula of solid will be AB.

BOTANY

91. Answer (1)

Vegetative phase precedes flowering event in a plant.

92. Answer (4)

The seeds germinate when favourable conditions for growth exist in the environment. In the absence of such favourable conditions the seeds do not germinate and goes into a period of suspended growth or rest.

93. Answer (2)

Growth is accompanied by metabolic processes (both anabolic and catabolic), that occurs at the expense of energy.

94. Answer (3)

Growth of a pollen tube is measured in terms of its length, an increase in surface area denotes the growth in a dorsiventral leaf.

95. Answer (3)

Meristematic are rich in protoplasm and possess large conspicuous nuclei. Their cell walls are primary in nature, thin and cellulosic with abundant plasmodesmatal connections.

The cells in the phase of maturation undergoes protoplasmic modification and have numerous vacuoles.

96. Answer (1)

Buttercup shows environmental heterophylly.

97. Answer (4)

Auxin is used as weedicide e.g. 2, 4-D.

98. Answer (3)

The given graph exhibits sigmoid growth curve.

99. Answer (4)

Auxin was isolated by F.W. Went from tips of coleoptiles of oat seedlings.

100. Answer (2)

Ethylene promotes root growth and root hair formation, thus helping the plants to increase their absorption surface.

101. Answer (3)

ABA are derivatives of carotenoids.

102. Answer (1)

The hormonal substance migrates from leaves to shoot apices for inducing flowering only when the plants are exposed to the necessary inductive photoperiod.

103. Answer (1)

(A) **Auxin:** Induce parthenocarpy in tomatoes.

(B) **Ethylene:** Promotes rapid internode/petiole elongation in deep water rice plants.

(C) **Abscisic acid:** Increases tolerance of plants to various kinds of stresses.

(D) **Gibberellins:** Promotes elongation and improvement of shapes in apples.

104. Answer (2)

The volatile substance was identified as ethylene, a gaseous PGR, released from ripened oranges that hastened the ripening of stored unripened bananas.

105. Answer (1)

Vernalisation refers specially to the promotion of flowering by a period of low temperature.

106. Answer (3)

Presence of chemical substances such as abscisic acids, phenolic acids, para-ascorbic acid causes seed dormancy.

Effect on inhibitory substances can be removed by application of certain chemicals like gibberellic acid and nitrates.

107. Answer (4)

Natural cytolinins are synthesized in the regions where rapid cell division occurs, for example, root apices, developing shoot buds, young fruits etc.

108. Answer (2)

Cytokinin promotes nutrient mobilisation which helps in the delay of senescence.

109. Answer (3)

In most situations, ABA acts as an antagonist to Gibberellins (GAs).

110. Answer (2)

Life-span of organisms are not necessarily correlated with their sizes.

111. Answer (1)

Life-span of -

Cow – 20-25 years

Horse – 60 years

Crow – 15 years

Parrot – 140 years

112. Answer (2)

There is no natural death in single-celled organisms, *e.g.*, *Amoeba, Paramoecium, Euglena,* etc.

Penicillium is a multicellular organism.

113. Answer (4)

Sexual reproduction produces genetic variations among offsprings while, clones are produced during asexual reproduction.

114. Answer (1)

The term clone refers to morphologically and genetically similar individuals. Clones are produced by asexual reproduction.

115. Answer (3)

Conidia are the asexual reproductive structures produced in *Penicillium*. Zoospores are reproductive structures produced in *Chlamydomonas.*

116. Answer (4)

Budding occurs in yeast in which division of cytoplasm is unequal.

117. Answer (1)

Vegetative propagules in various angiosperms are:

(A) Ginger – Rhizome

(B) *Agave* – Bulbil

(C) *Bryophyllum* – Leaf buds

(D) Water hyacinth – Offset

118. Answer (2)

Eichhornia (or, water hyacinth) is known as 'terror of Bengal'. Its short horizontal branch producing a cluster of leaves above and cluster of roots below is called offset.

119. Answer (2)

In banana vegetative propagation occurs through rhizome.

120. Answer (1)

Algae produce isogametes as well as heterogametes.

121. Answer (1)

Polycarpic plants bear flower throughout the life, *e.g.* China rose (shoe flower).

122. Answer (4)

Most of the species of *Chara* are monoecious. These have male sex organ (globule), located below nucule (female sex organ).

123. Answer (2)

When the fusing gametes are morphologically similar, they are known as isogametes. e.g. *Ulothrix.*

124. Answer (1)

In most of the algae, syngamy occurs outside the body in external medium

125. Answer (3)

Zoospores produced in algae is asexual structure, motile, may be haploid or diploid and usually naked.

126. Answer (1)

Embryogenesis is absent in algae.

127. Answer (1)

In flowering plants, fertilized ovule matures into a seed.

128. Answer (2)

According to given question,

Number of chromosomes in meiocyte of individual = 78

$\therefore$ Gamete chromosome

$= \frac{1}{2} \times$ Meiocyte chromosome

$= \frac{1}{2} \times 78 = 39$

129. Answer (4)

Recovery phase is seen in life-cycle of polycarpic plants, *e.g.*, Jackfruit, Mango, etc.

130. Answer (1)

Marchantia is a dioecious bryophyte. Rest of other statements are true.

131. Answer (2)

Cytokinin helps to produce chloroplasts in leaves.

132. Answer (1)

The site of perception of light/dark duration is leaves.

133. Answer (1)

Auxins was first isolated from human urine.

134. Answer (2)

IBA (Indole-3-butyric acid) is a natural auxin.

135. Answer (3)

Ethylene decreases sensitivity of roots to gravity.

ZOOLOGY

136. Answer (1)

Organs influenced by a hormone are called its target organs which contain receptors for that particular hormone. Hormones are considered as intercellular messengers.

137. Answer (4)

Hormones are non-nutrient chemicals synthesized in traces and act as chemical messengers. They may be proteins, polypeptides, steroids and biogenic amines in chemical nature.

138. Answer (4)

Hormones are poured directly into interstitial fluid bathing the gland.

139. Answer (3)

Although neural system is faster than endocrine system, but all cells of the body are not innervated by nerve fibres. So both systems jointly coordinate and regulate physiological functions of the body.

140. Answer (1)

Both thyroid and parathyroid are organised endocrine glands. Liver and heart are not organised endocrine glands. Testes and pancreas are mixed-composite glands.

141. Answer (3)

Aldosterone is steroid hormone secreted from adrenal cortex.

142. Answer (3)

Sex hormones, corticosteroids and hormones of thyroid gland (T_4 and T_3) act by altering gene expression.

143. Answer (2)

Neurogenic hormones of hypothalamus regulate secretion of hormones from anterior lobe of pituitary gland.

144. Answer (4)

Cortisol is cholesterol derivative hormone and is steroid in its chemical nature.

145. Answer (3)

Pineal gland secretes hormone melatonin which regulates normal rhythms of sleep wake cycle and influences defence capability of our body. Thyrocalcitonin decreases blood calcium level. Gonadotrophins are hormones of anterior pituitary. Somatostatin is secreted from hypothalamus.

146 Answer (4)

Protrusion of eyeball occurs in Graves' disease.

147. Answer (3)

Melatonin does not influences erythropoiesis.

148. Answer (4)

Parathormone is under control of circulating level of Ca^{+2} in blood.

149. Answer (1)

Hyposecretion of PTH (Parathormone) leads to deficiency of Ca^{+2} in blood which leads to tetany.

150. Answer (1)

Adenohypophysis consists of two parts-pars distalis and pars intermedia. Pars distalis is commonly called anterior pituitary.

151. Answer (1)

A $\rightarrow$ Hypothalamic neurons

B $\rightarrow$ Portal circulation

C $\rightarrow$ Posterior pituitary

D $\rightarrow$ Anterior pituitary

152. Answer (3)

Adrenaline increases blood supply of vital organs such as brain and heart, causes piloerection and increases blood glucose level by glycogenolysis.

153. Answer (2)

Insulin hormone is water soluble and cannot cross plasma membrane. It acts by forming hormone-receptor complex with receptor present on cell surface.

154. Answer (3)

Hypothalamus synthesises ADH which is stored and released from posterior pituitary.

155. Answer (4)

Adrenal cortex synthesises aldosterone and thyroid gland synthesises thyroxine. PTH increases blood calcium level leading to hypercalcemia.

156. Answer (4)

Asexual reproduction involves single parent.

157. Answer (2)

Encystation in *Amoeba* occurs only during unfavourable condition. Sexual reproduction is absent in *Amoeba*.

158. Answer (1)

Germ cell is a diploid cell whereas other cells given in options are haploid.

159. Answer (4)

Life cycle is events occurring during life span of an organism. Life expectancy is based on average life span.

160. Answer (4)

Hydra undergo budding during favourable condition. *Euglena* perform longitudinal binary fission. Simple irregular binary fission occurs in *Amoeba.*

161. Answer (2)

Birds have internal fertilization and external development.

162. Answer (3)

Butterfly – 1-2 weeks

Crow – 15 years

Crocodile – 60 years

163. Answer (4)

In unicellular organisms, cell division is itself a mode of reproduction.

164. Answer (1)

Fusion of gametes occurs during fertilization. Embryogenesis is included in post-fertilization events.

165. Answer (1)

A haploid parent produces gamete by mitosis.

166. Answer (2)

Meiocytes are diploid (2n) cells and gametes are haploid cells.

167. Answer (4)

In parthenogenesis, female's egg undergoes development which may be haploid or diploid. Zygote is produced only in sexual reproduction which is always diploid.

168. Answer (1)

Zygote is a vital link between organism of one generation and the next.

169. Answer (3)

Hormones are responsible for transitions between the three phases of life e.g. juvenile/vegetative phase, reproductive phase and senescence phase.

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170. Answer (2)

Mineralocorticoids are secreted by feedback mechanism and Na^+ and K^+ concentration in the body.

171. Answer (3)

Development of mammary glands in females and male is function of oestrogen. In males, development of mammary glands is called gynaecomastia.

172. Answer (4)

Pituitary gland secretes 6 hormones from its adenohypophysis, only one hormone from pars intermedia and 2 hormones from pars nervosa.

173. Answer (1)

Thyroxine regulates development and maturation of CNS, so called intelligence hormone. Thymosins regulate both cell mediated and humoral immunity.

174. Answer (4)

Glycogenolysis is function of catecholamines.

175. Answer (2)

ANF decreases blood pressure by vasodilation. CCK acts on pancreatic acini to increase secretion of enzymes in pancreatic juice. Progesterone maintains pregnancy by increasing blood supply to uterus.

176. Answer (1)

Aldosterone is a steroid hormone and forms hormone receptor complex within cell to interact with genome.

177. Answer (4)

Acromegaly is caused due to hypersecretion of GH in adults.

178. Answer (4)

Primary sex organs are formed under control of genes.

179. Answer (4)

FSH acts on cells of Sertoli to secrete some factor which help in spermatogenesis.

180. Answer (2)

Thymus gland is degenerated in old persons resulting in decrease production of thymosins. As a result, immune responses of old persons become weak.

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